Structure-Function Integration to Identify Lesions

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College of Osteopathic Medicine

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- Department of Biomedical and Anatomical Sciences

Parasaurolophus in New Mexico Musuem of Natural History

National Senior Olympics

Session Objectives

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- Review the CNS blood supplies.
- Differentiate the major functions of different lobes.
- Describe the major sensory & motor functional systems.
- Master the cerebellar pathways.
- Integrate the structure – function to identify lesions.

Before we start….

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- Prepare pens and papers.
- Be sure to pause the video frequently and allow yourself enough time to draw out the pathways!

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Major blood supply for CNS

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Two major arteries and their branches supply the brain and spinal cord:

- Vertebral a. ____________________ for **posterior** circulation
- Internal carotid a. ____________________ for **anterior** circulation

Please take out a piece of scratch paper and draw the Circle of Willis (including vertebral artery and its major branches)

Be sure to label all branches of blood vessels (abbreviations OK)

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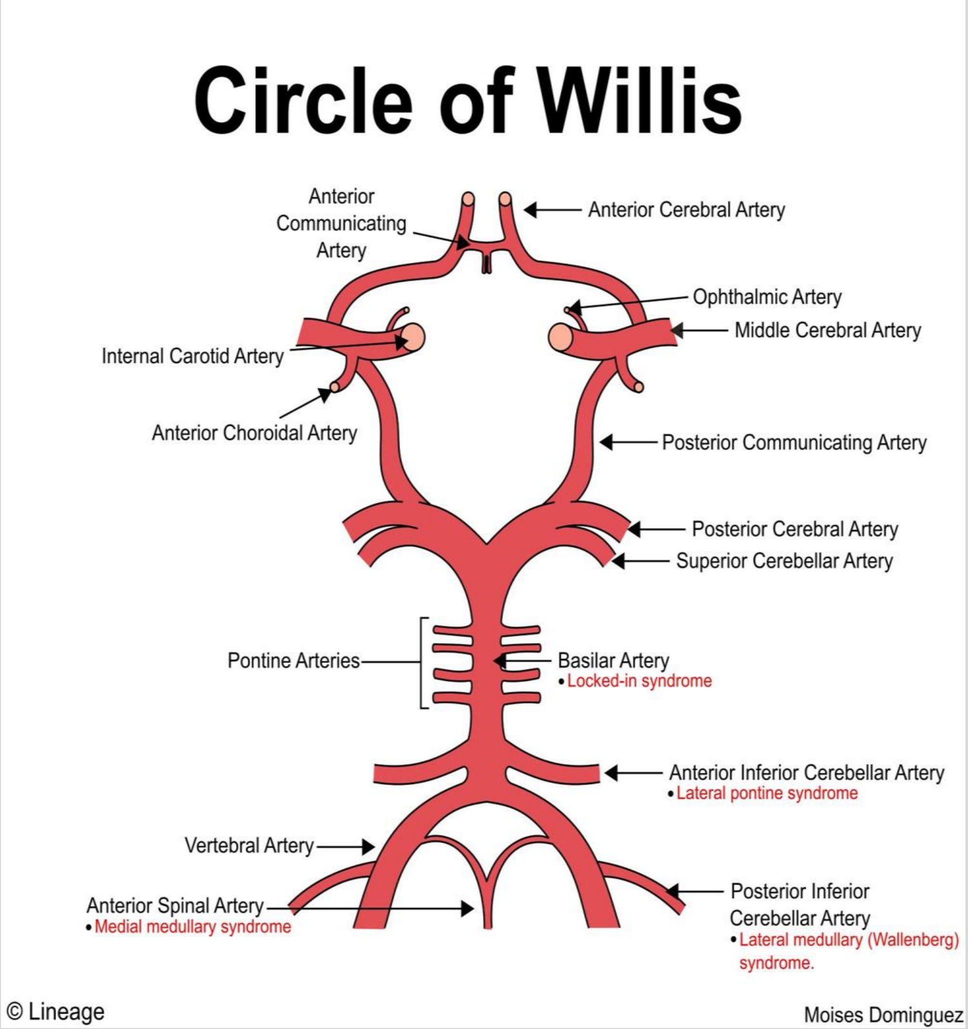
Pause here…..



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https://step1.medbullets.com/neurology/113022/circle-of-willis

Challenge:

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Can you trace back the CNS blood supply to the aorta?

Pause here…..

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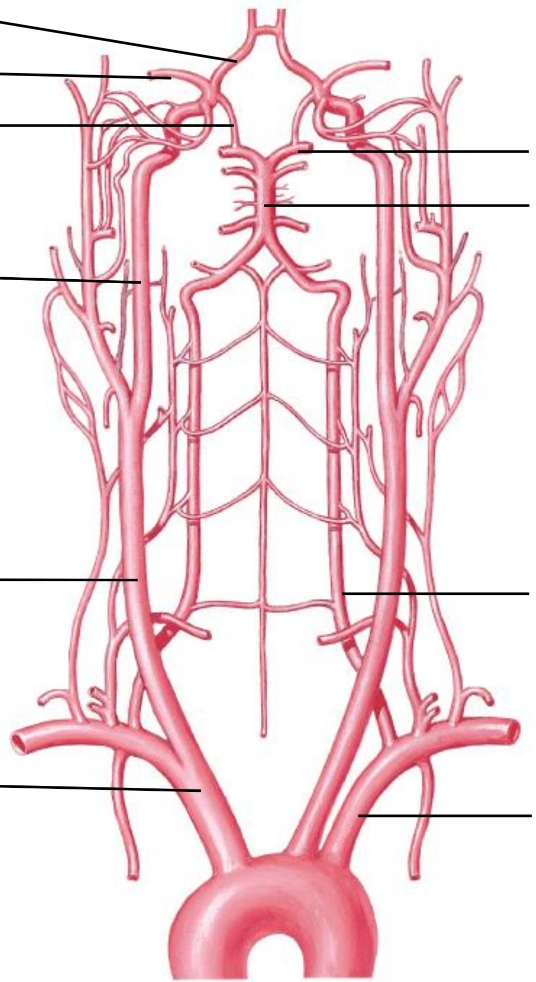
Anterior supply

Arteries to Brain: Schema

Posterior supply



Lateral view

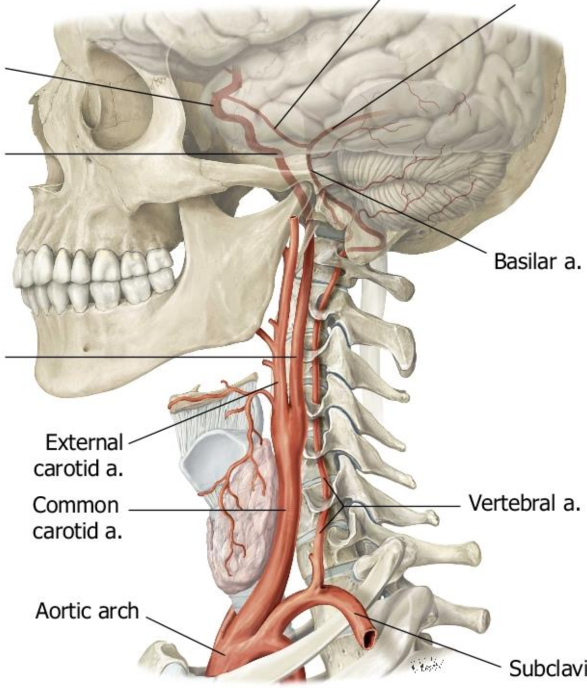


Fig. 41.4 (Illustrator: Karl Wesker)

Now add the brain underneath the anterior and posterior circulation- how are you going to line them up?

Hint: Think about the relative position of these vessels to the brain subdivisions…

See next slide

Where would you position these lines?

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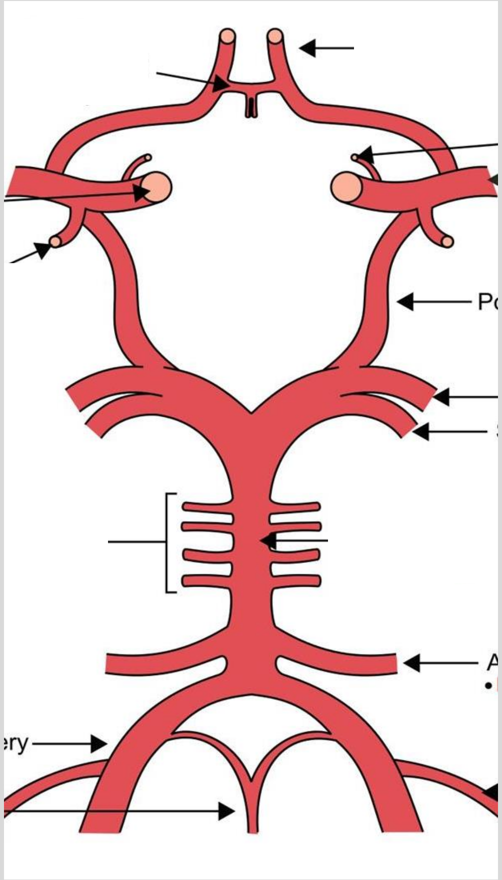
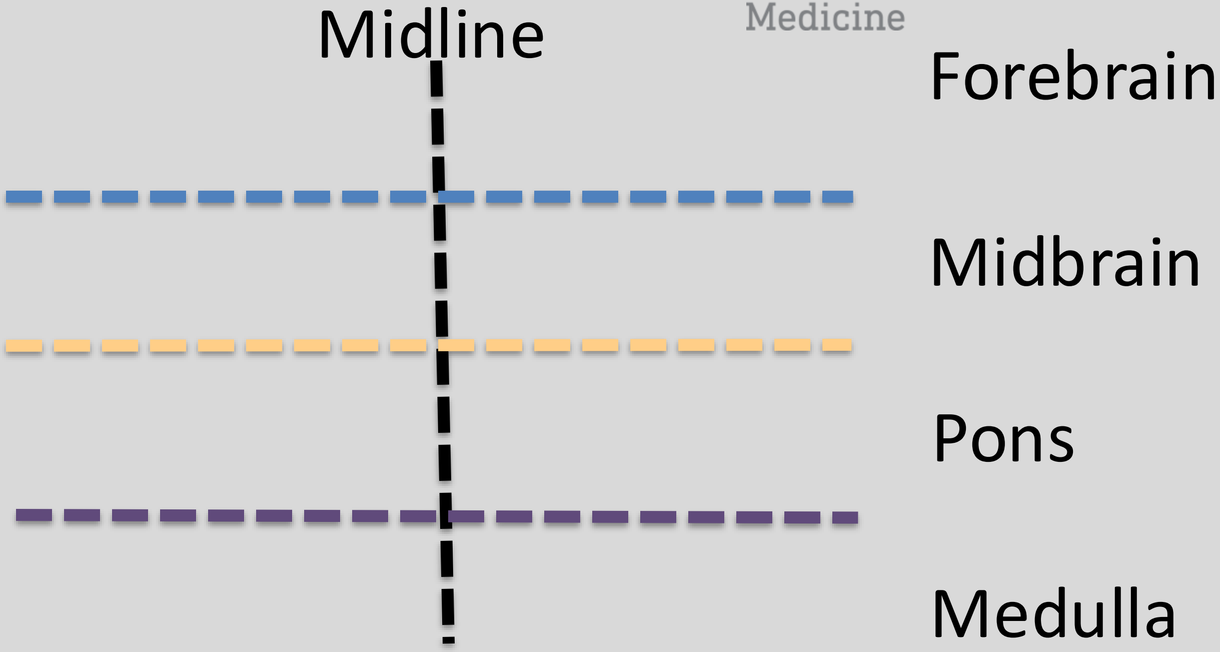
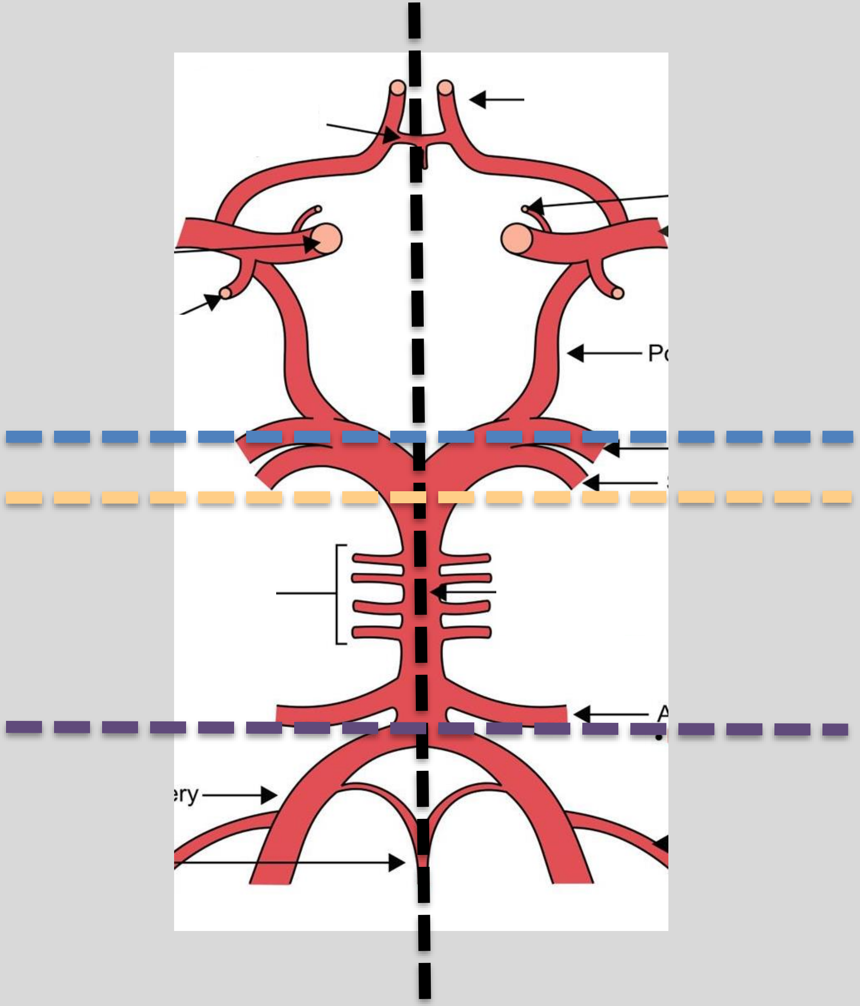




Pause here…..

Midline



Forebrain

Midbrain

Pons

Medulla

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Next, let's add the cranial nerves. Please label the locations where the CNs come out of the brain.

Hint: Think about the relative position of these nerves to different parts of the brainstem… Which segment? Medial or lateral?

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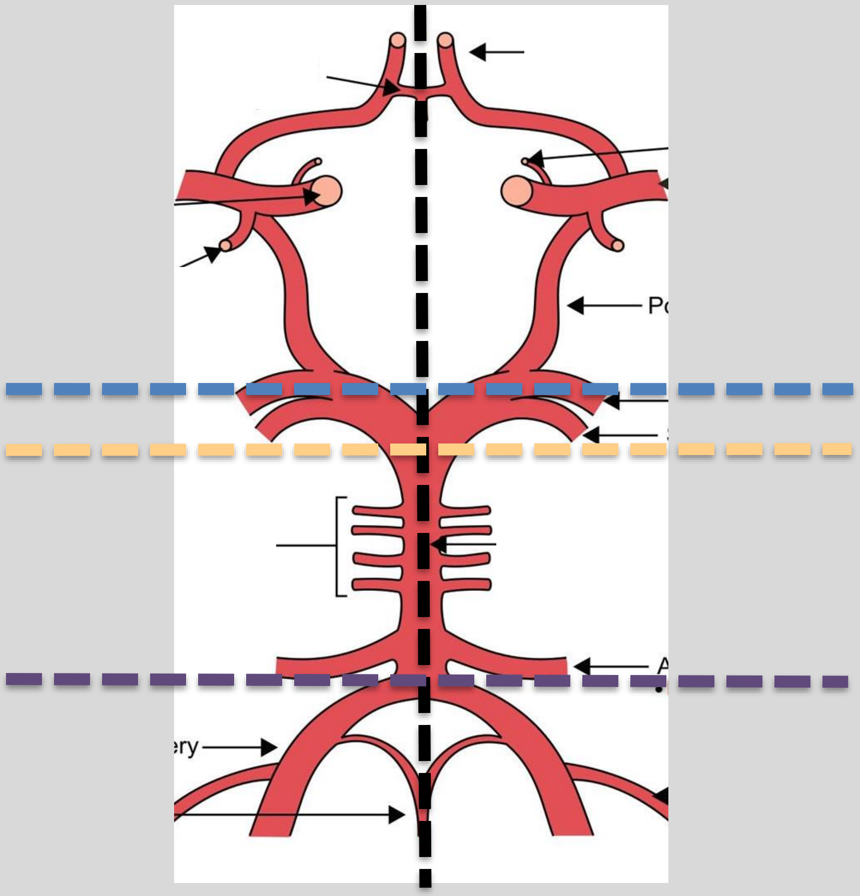
See next slide

Midline



Forebrain

Midbrain

Pons

Medulla

1
2
3
4
5
6
7
8
9
10
11
12

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Pause here.....

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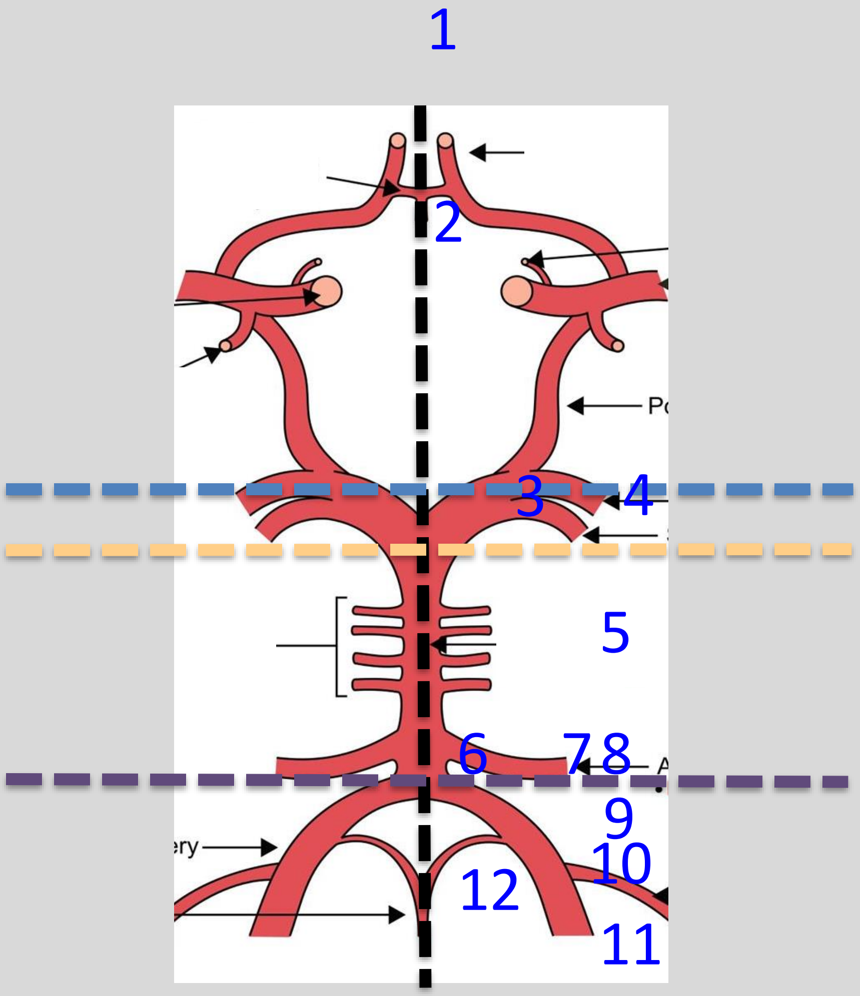


Forebrain

Midbrain

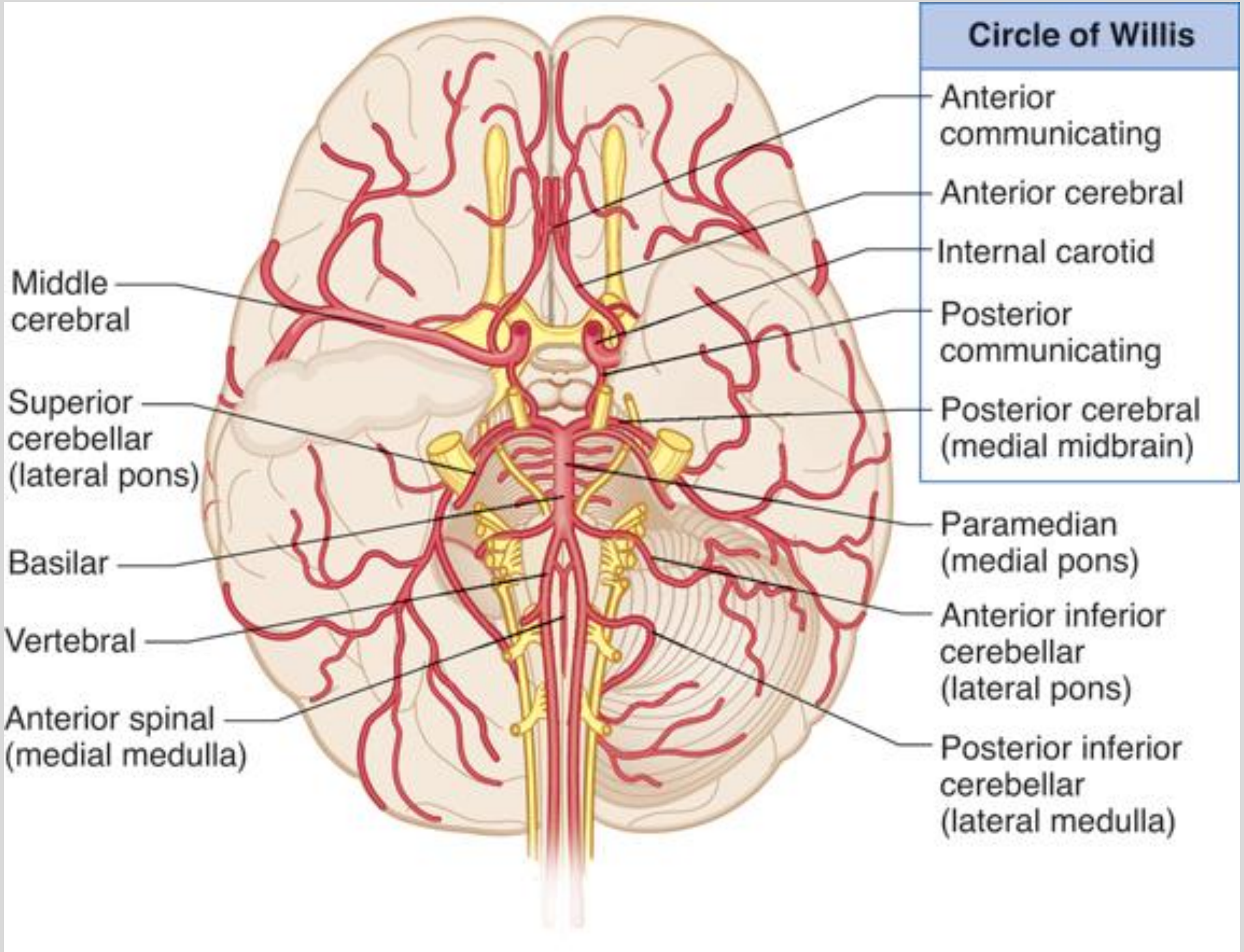
Pons

Medulla

"Rule of 4s"

Gates

Gates. Int Med J 2005; 35: 263-6
https://onlinelibrary.wiley.com/doi/epdf/10.1111/j.1445-5994.2004.00732.x



https://www.pinterest.com/pin/293296994463055105/

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Major Structures & Functions

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1)

2)

3)

4)

5)

6)

7)

8)

Pause here.....

Nolte's the Human Brain in Photographs and Diagrams, 5th ed.

Major Structures & Functions

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1) Pre-motor
- Planning movements
- Frontal eye field

2) Primary gustatory
(anterior insula & frontal operculum)

3) Primary Auditory - Both ears

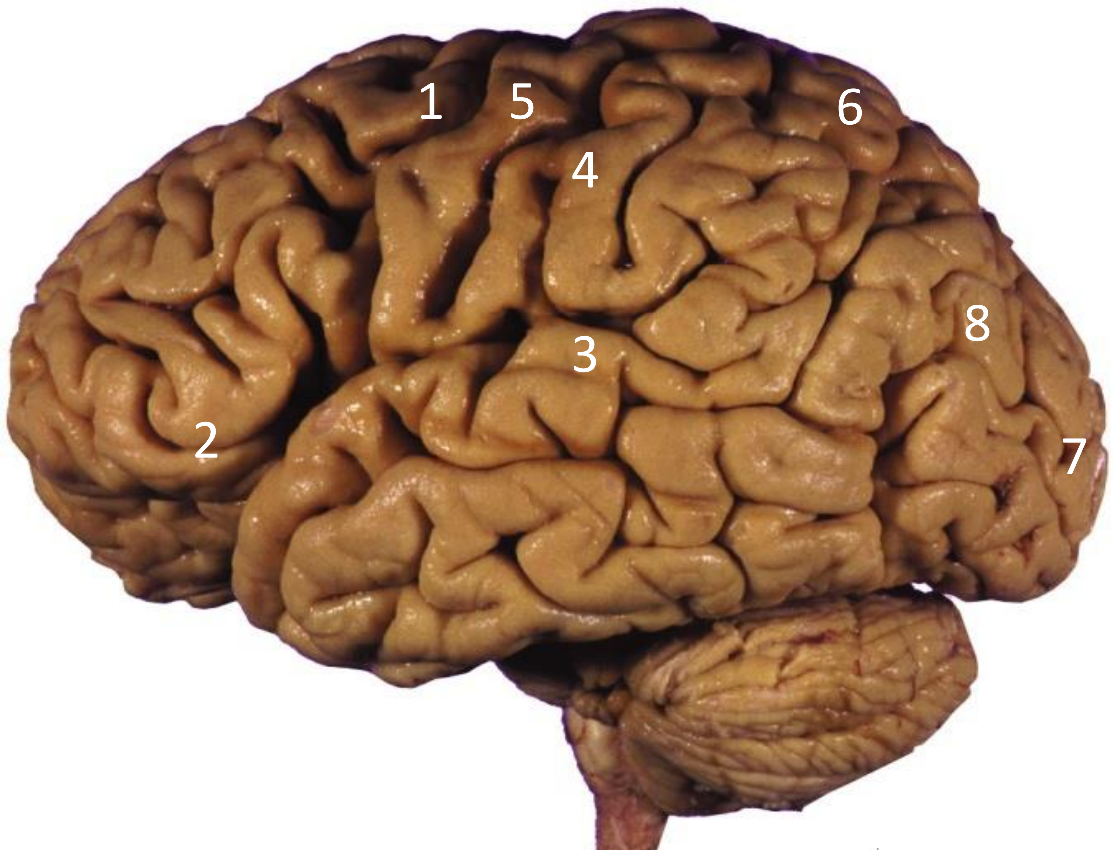
4) Primary Somatosensory

5) Primary Motor

6) Somatosensory Association

7) Primary Visual

8) Visual Association
- Motion & Location



Nolte's the Human Brain in Photographs and Diagrams, 5th ed.

Major Structures & Functions

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1)

2)

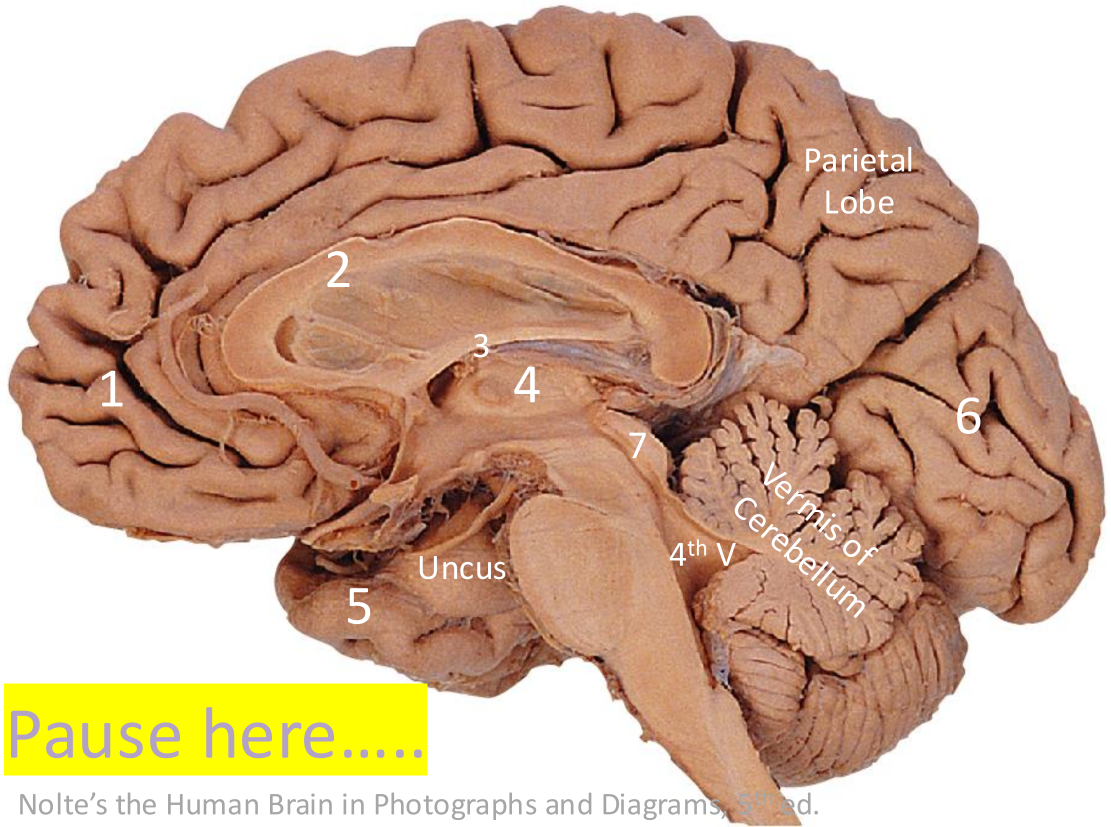
3)

4)

5)

6)

7)



Pause here…..

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Major Structures & Functions

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1) Pre-frontal cortex
   - Personality
   - Executive functions

2) Corpus callosum
   - Cerebral hemisphere communications

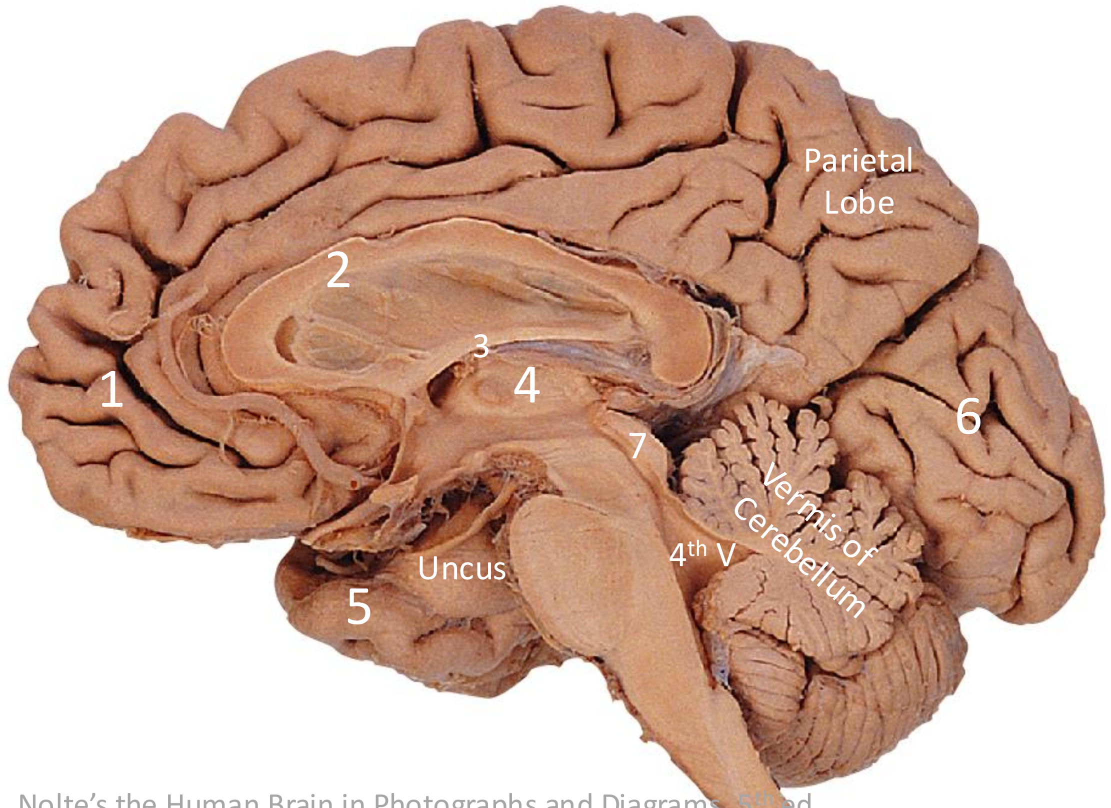
3) Fornix
   - Output of the hippocampus

4) Thalamus
   - Sensory relay & association

5) Piriform cortex
   - Primary olfactory cortex

6) Calcarine sulcus
   - Primary visual cortex

7) Tectum
   - Superior colliculus -vision
   - Inferior colliculus -auditory



Nolte's the Human Brain in Photographs and Diagrams, 5th ed.

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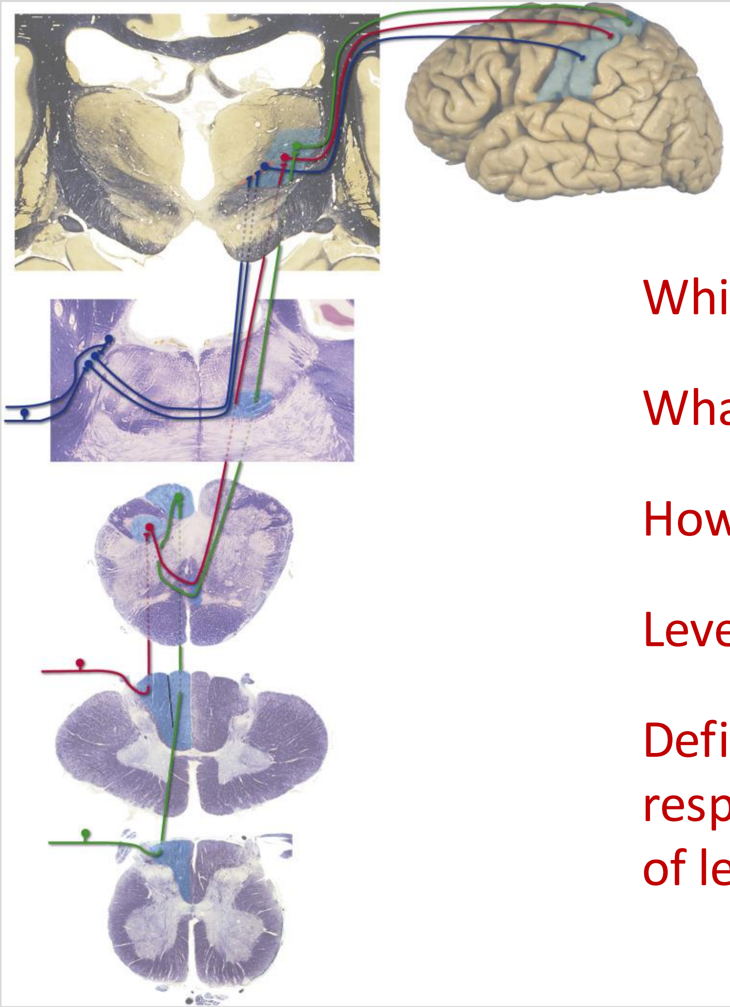
Long Tracts



Which tract?

What modality?

How many neurons relay?

Level of decussation?

Deficits lateralization in respect to different levels of lesions?

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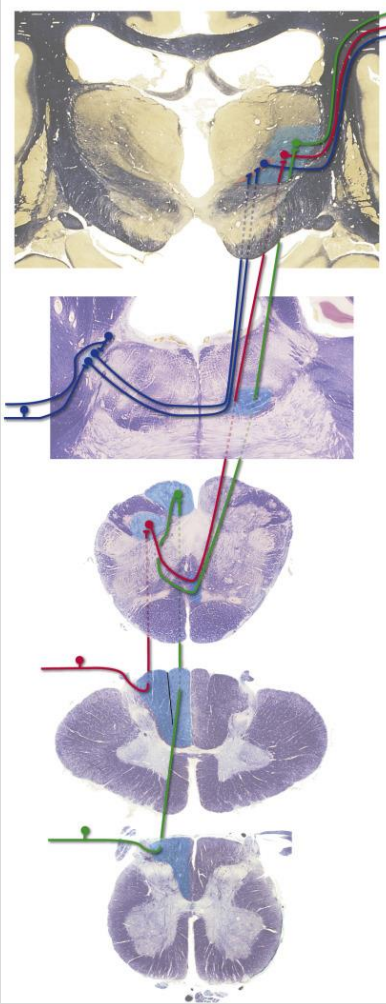
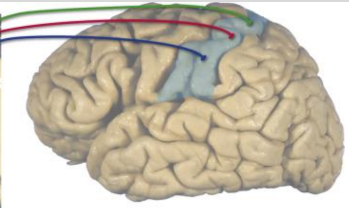
Pause here…..

Ascending

Dorsal Column – Medial Lemniscus (DCML) Pathway





Which tract? - DCML

What modality?

– Sensory, touch/proprioception

How many neurons relay?

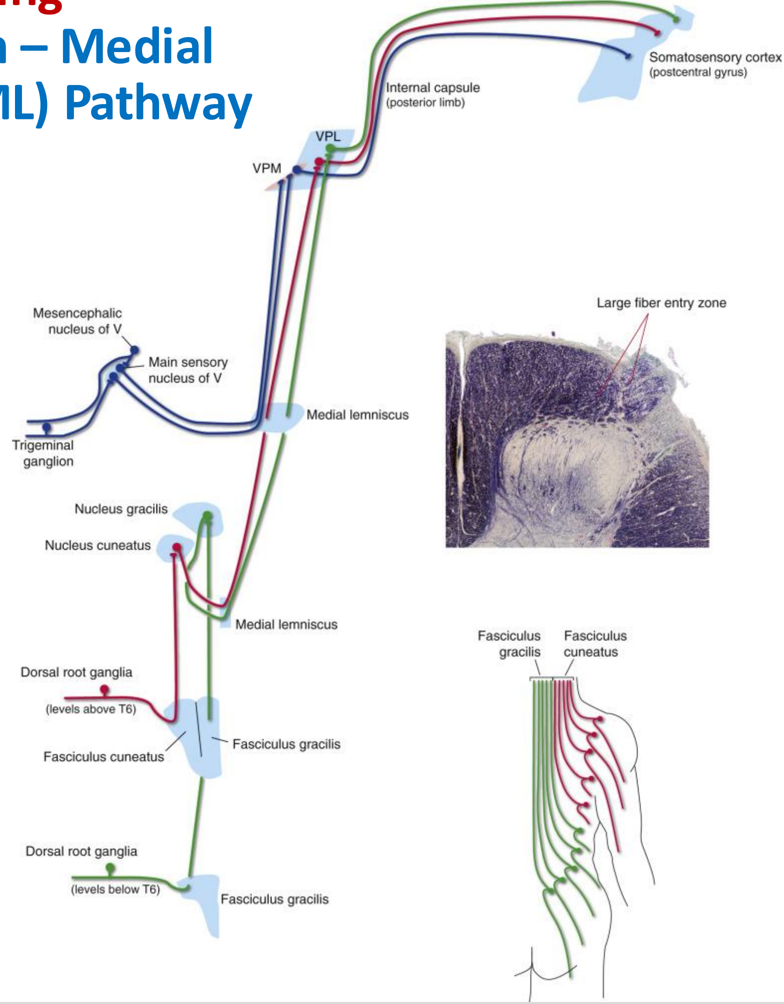
– 3 neurons: 1st order (primary afferents, DRG), 2nd order (medulla), 3rd order (VPL)

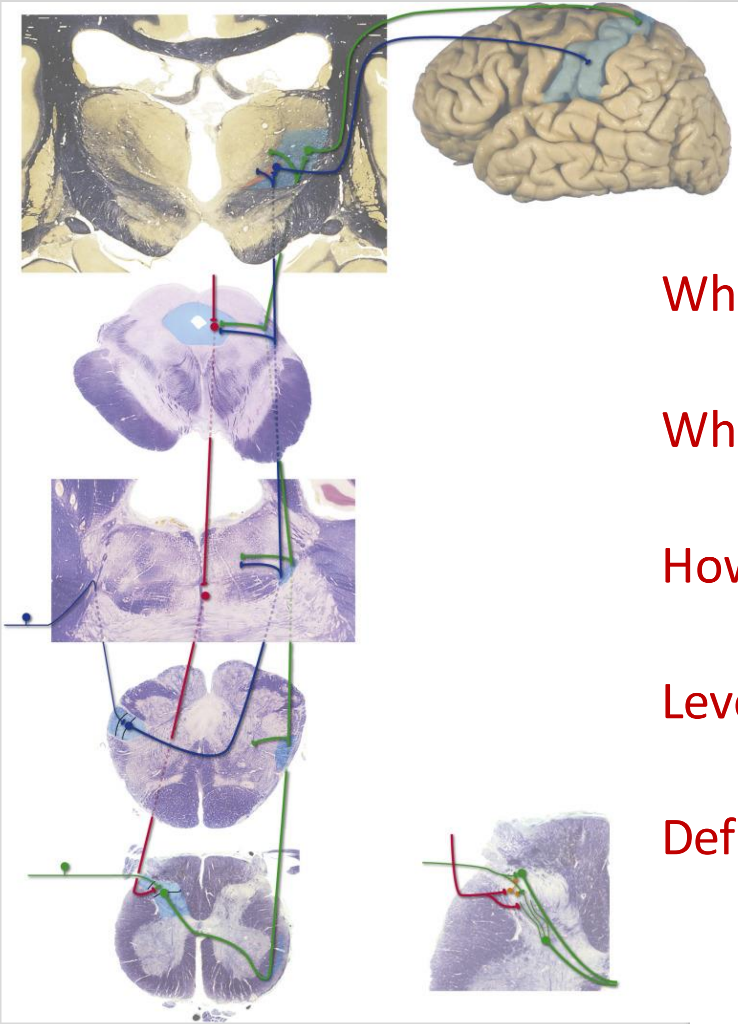
Level of decussation?

– Caudal medulla (n. gracilis & cuneatus)

Deficits lateralization?

– Contralateral to the lesion when above medulla; ipsilateral to the lesion when below medulla (spinal cord)





Which tract?

What modality?

How many neurons relay?

Level of decussation?

Deficits lateralization?

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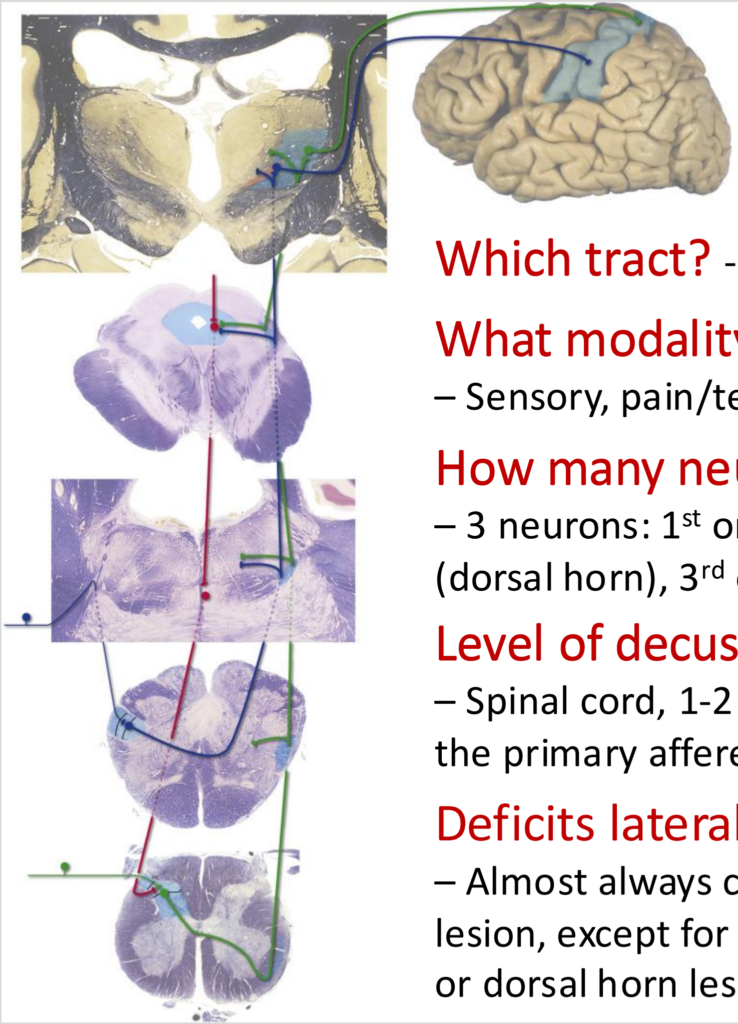
Pause here.....

Ascending
Spinothalamic Tract



Which tract? - STT

What modality?

– Sensory, pain/temperature

How many neurons relay?

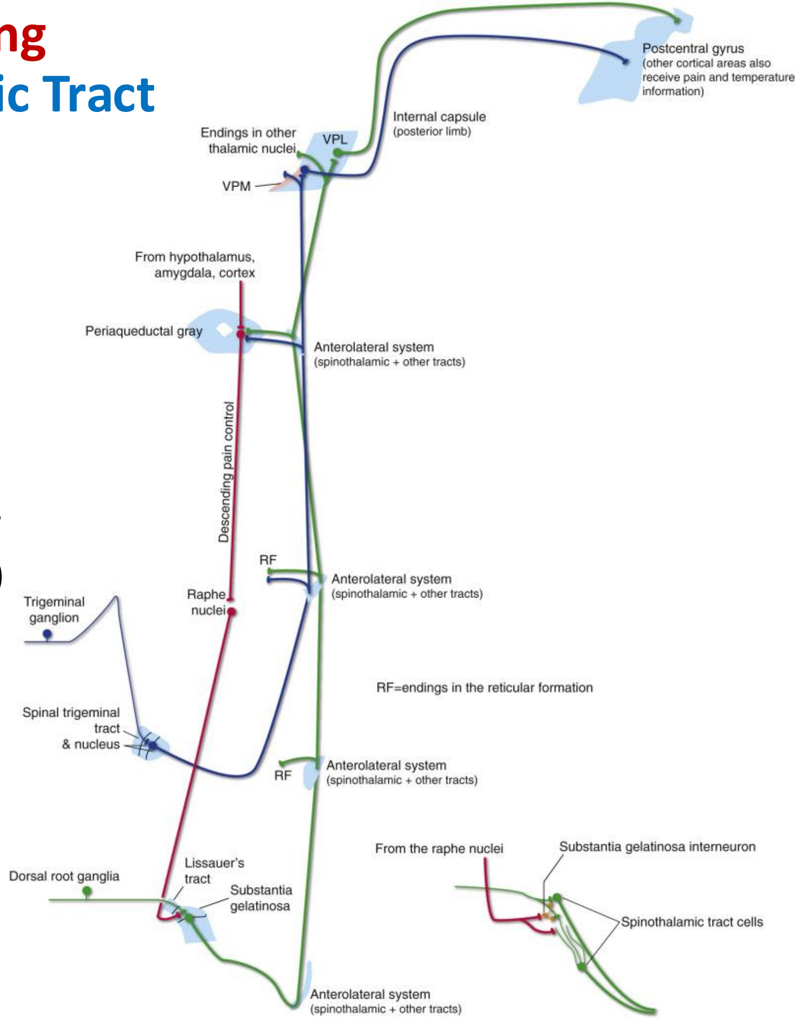
– 3 neurons: 1st order (DRG), 2nd order (dorsal horn), 3rd order (thalamus VPL)

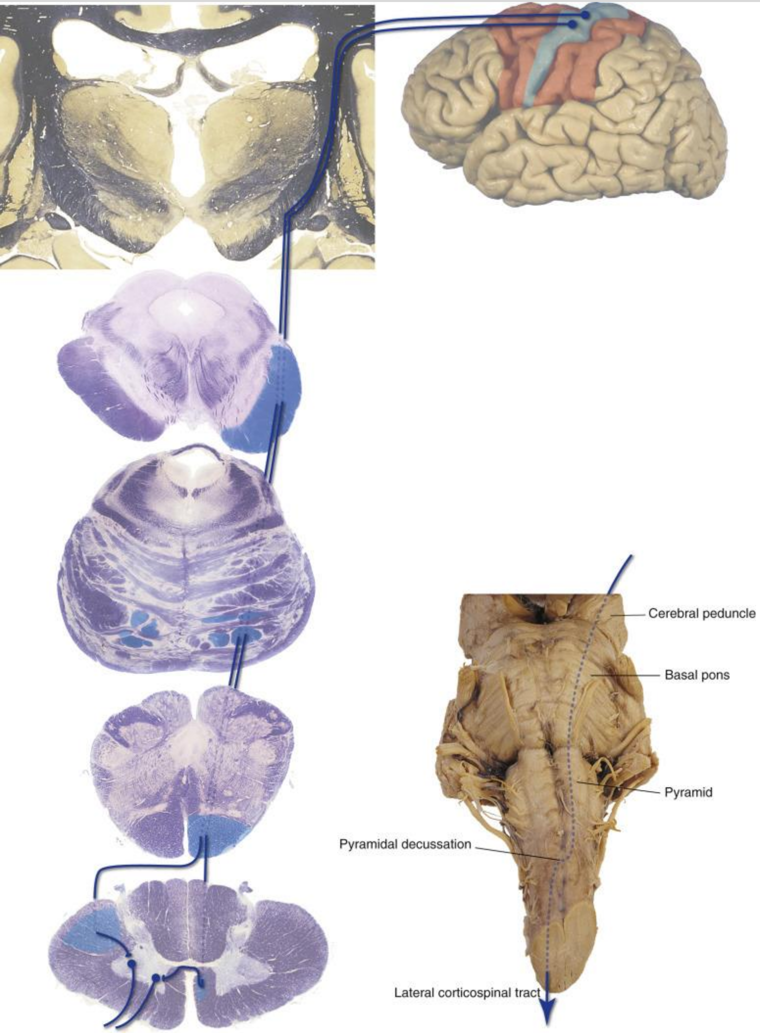
Level of decussation?

– Spinal cord, 1-2 segments above the primary afferents level

Deficits lateralization?

– Almost always contralateral to the lesion, except for primary afferents or dorsal horn lesions





Which tract?

What modality?

How many neurons relay?

Level of decussation?

Deficits lateralization?

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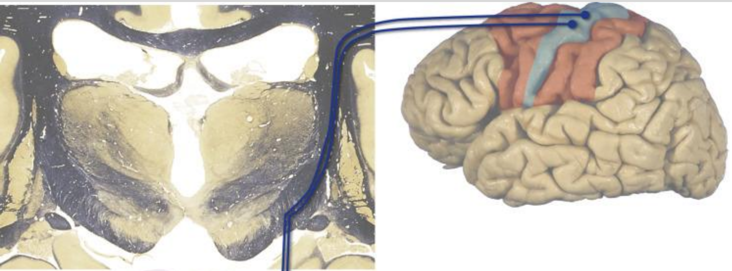
Pause here…..

Descending
Corticospinal Tract (pyramidal tract)



Which tract? - CST

What modality?

– Motor, voluntary movement

How many neurons relay?

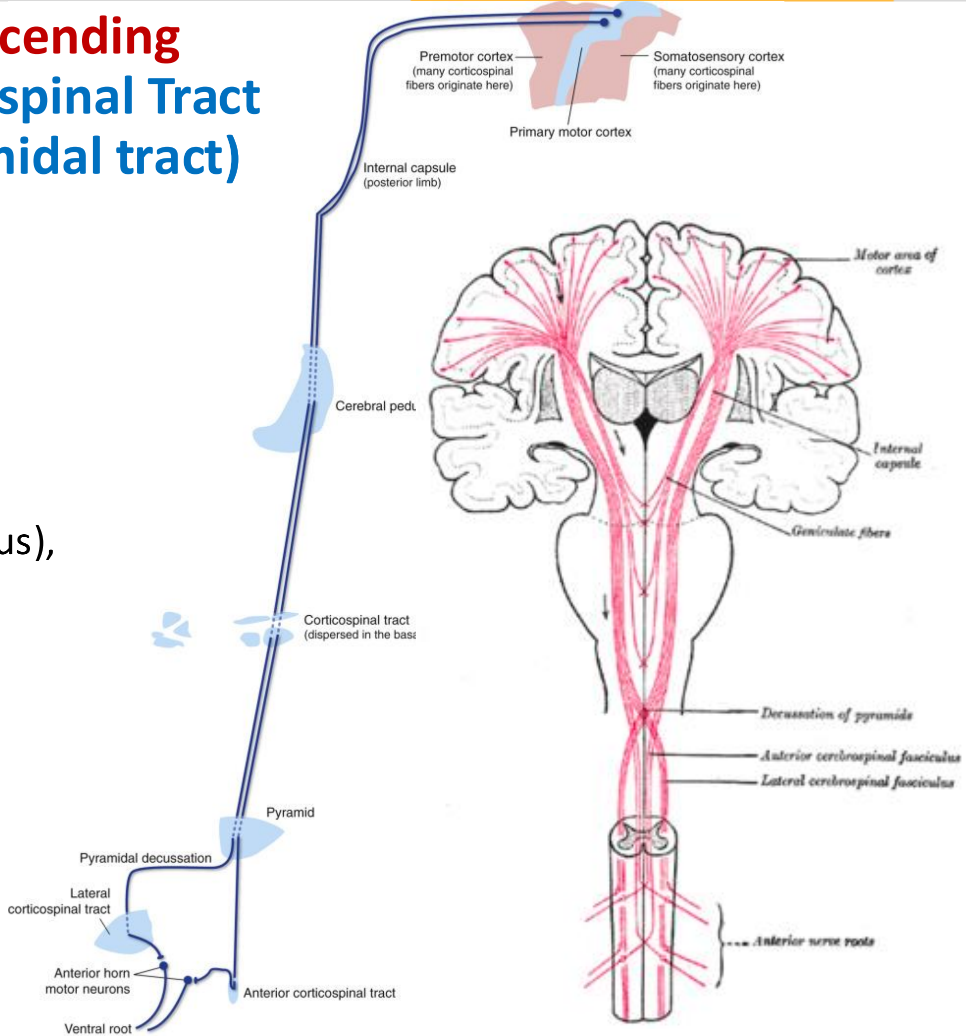
– 2 neurons: UMN (precentral gyrus), LMN (ventral horn)

Level of decussation?

– Pyramidal decussation (junction between medulla and spinal cord)

Deficits lateralization?

– Contralateral to the lesion above the spinal cord; ipsilateral to the lesion at the spinal cord



Fill in the chart:

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Lesions

Deficits

		UMN	LMN
Cortico spinal tract	Contraction		
	Reflex		
	Babinski sign		
	Lateralization		
Cortico bulbar tract	Contraction		
	Reflex		
	Lateralization		

Increase or decrease?

Exceptions for lateralization?

Pause here…..

Lesions

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Deficits

		UMN	LMN
Cortico-spinal tract	Contraction	↑	↓
	Reflex	↑	↓
	Babinski sign	**Y**	**N**
	Lateralization	ipsi/ contra	ipsi

		UMN	LMN
Cortico-bulbar tract	Contraction	**None**	↓
	Reflex	**None**	↓
	Lateralization	**NA**	ipsi

Exceptions (cortico-bulbar):

UMN: Lower face is innervated by contralateral UMNs.

LMN: Eye muscle innervations nucleus vs. nerve difference

Eye Movement – CN III, IV, VI

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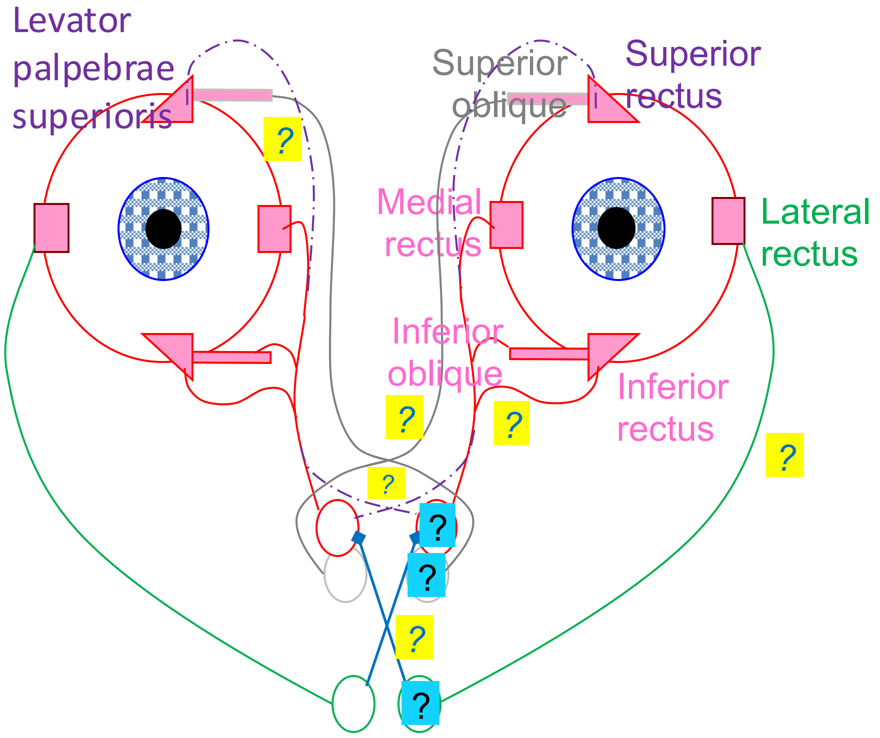

Which nerve/ fiber connection? Function?

Which nucleus? Function?

Presentation differences of nerve vs. nucleus damage?

Pause here…..

Eye Movement – CN III, IV, VI

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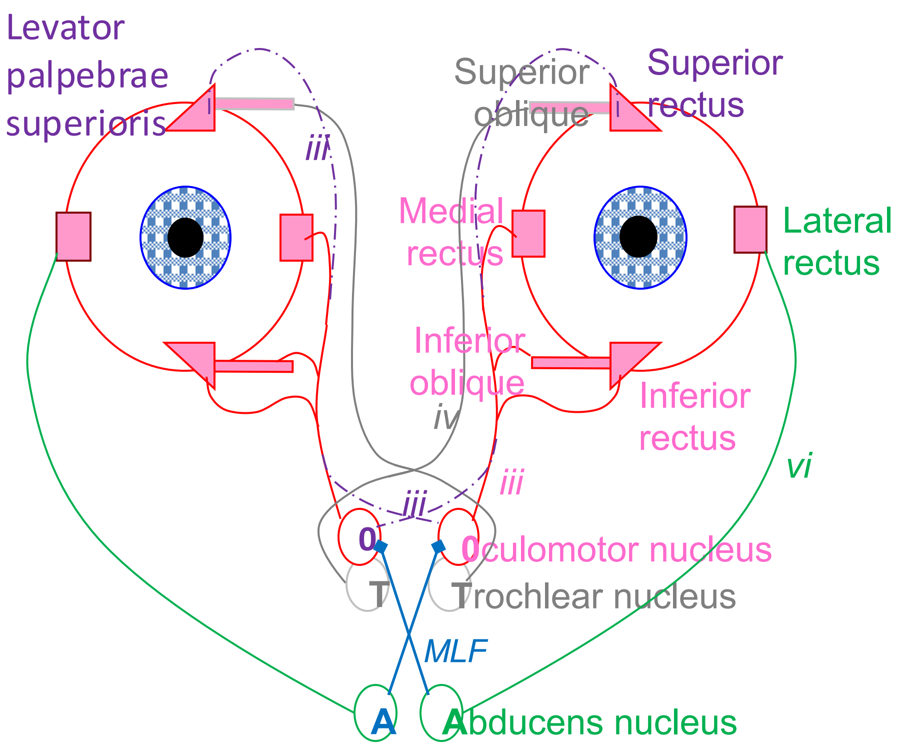


CN III (oculomotor):

- inferior division (MR, IR, IO)
  + EW nucleus (parasympathetic) → bilateral pupil sphincter & ciliary muscles
- superior division (SR & LPS – from contralateral oculomotor nucleus)

CN IV (trochlear):

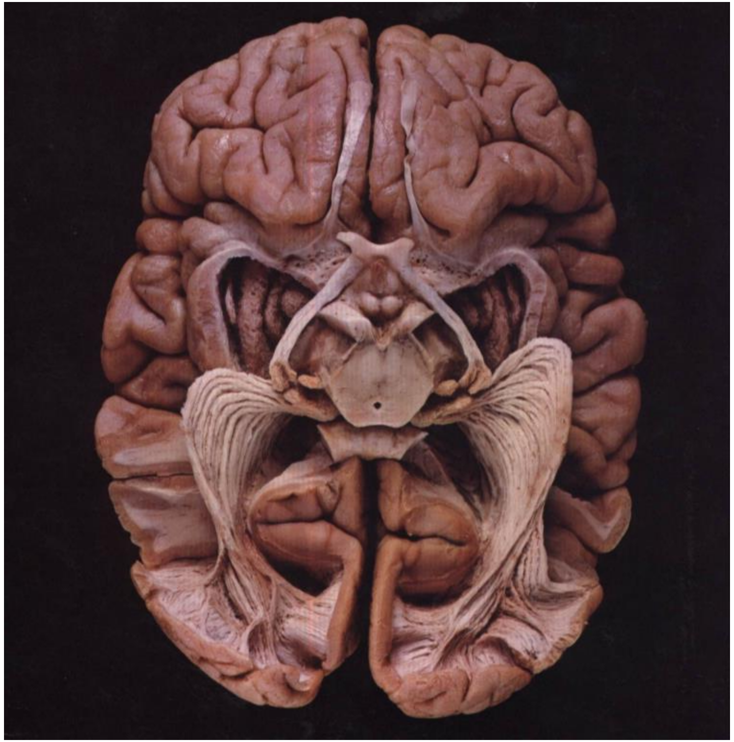
- SO – from contralateral trochlear nucleus

CN VI (abducens):

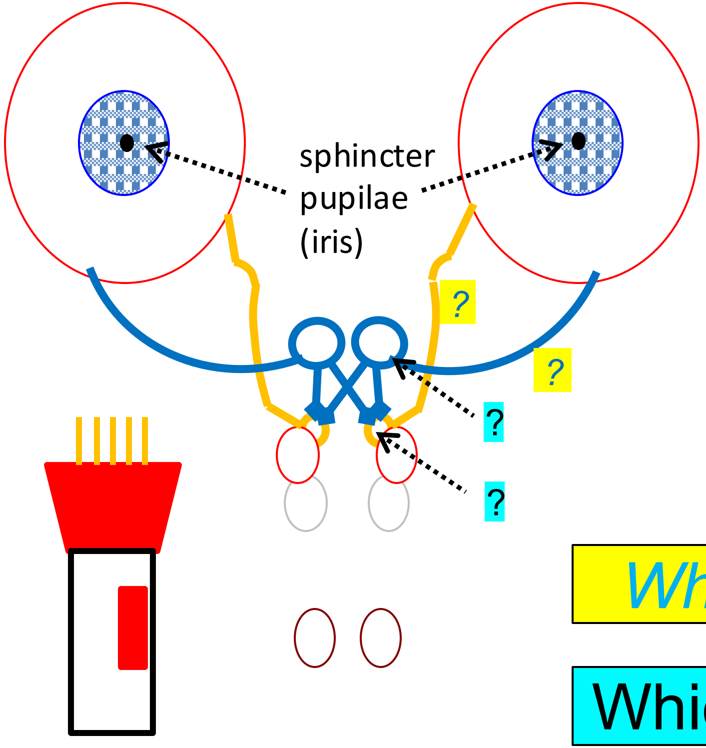
- LR
- Abducens nucleus also activates contralateral oculomotor nucleus via MLF (medial longitudinal fasciculus)

Dr. Kurtzer's lecture

Light Reflex



retina → ________ → ___________ → _____________ → **each** CN III → **each** iris

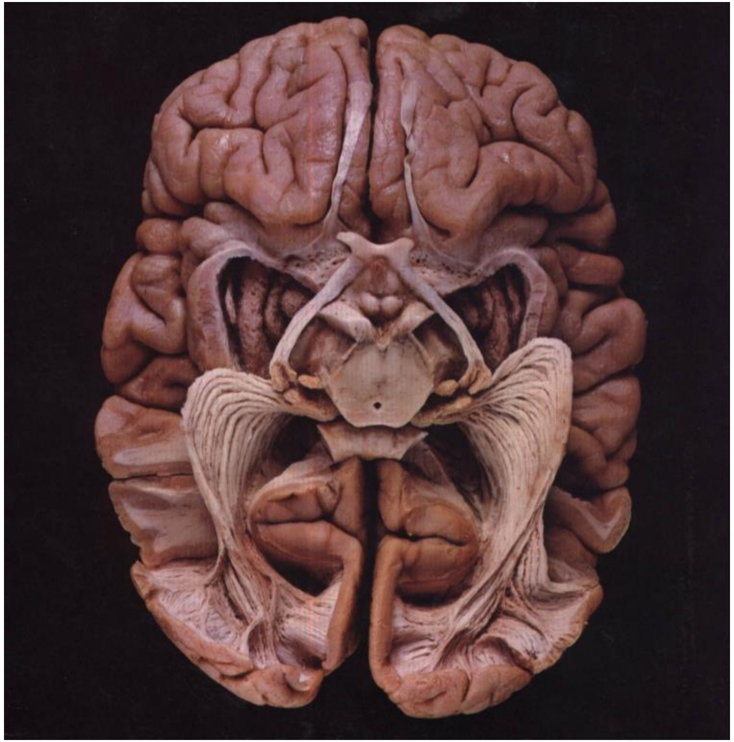


Which nerve?

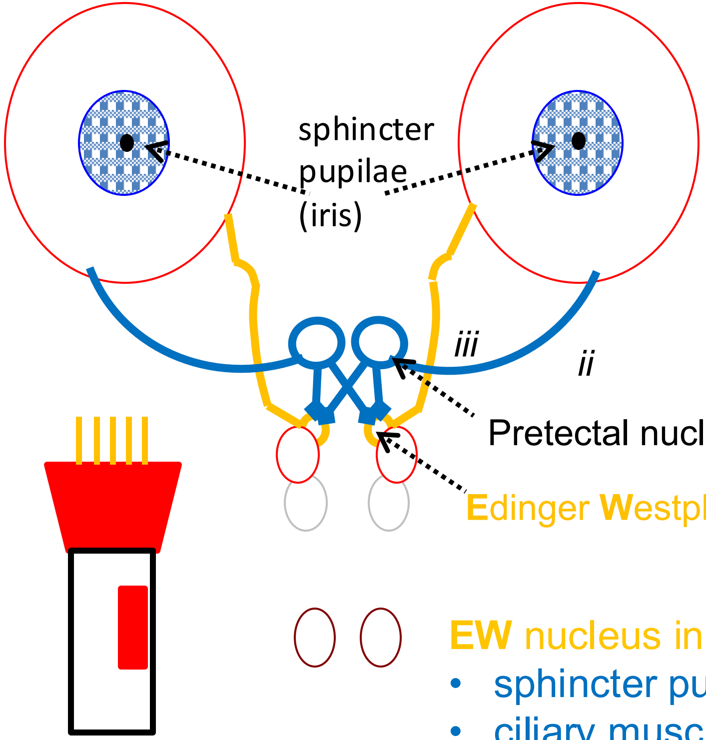
Which nucleus?

Pause here…..

Light Reflex



retina → CN II → Pretectal nucleus → **Bilateral** EW nucleus → **each** CN III → **each** iris



EW nucleus innervates

- sphincter pupillae (pupil)
- ciliary muscle (accommodation)

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Cerebellar Structures

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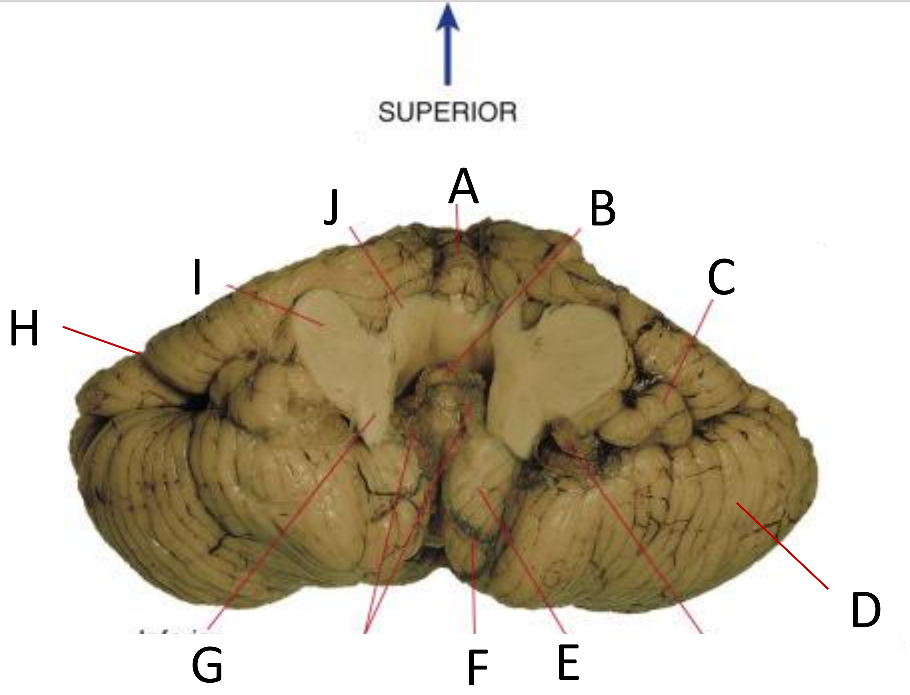
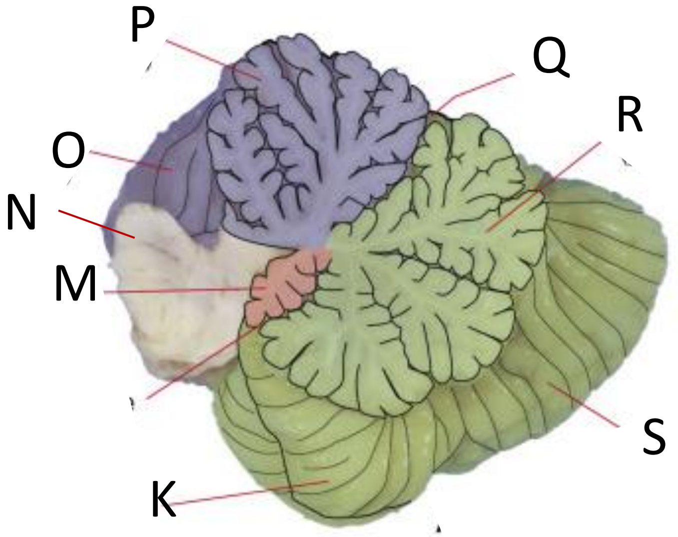
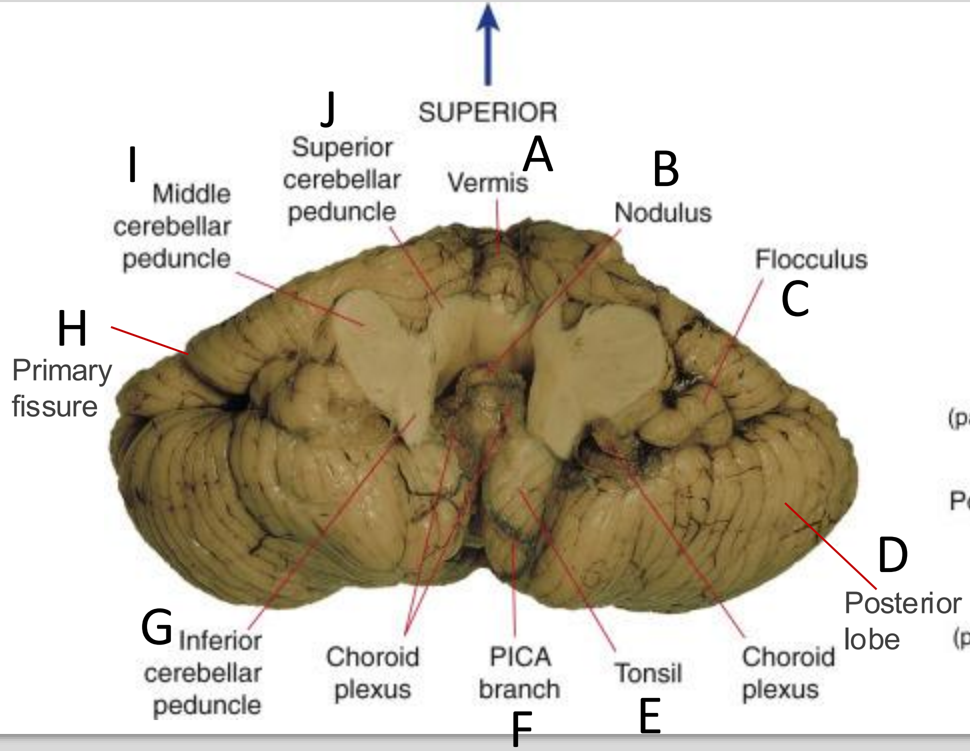


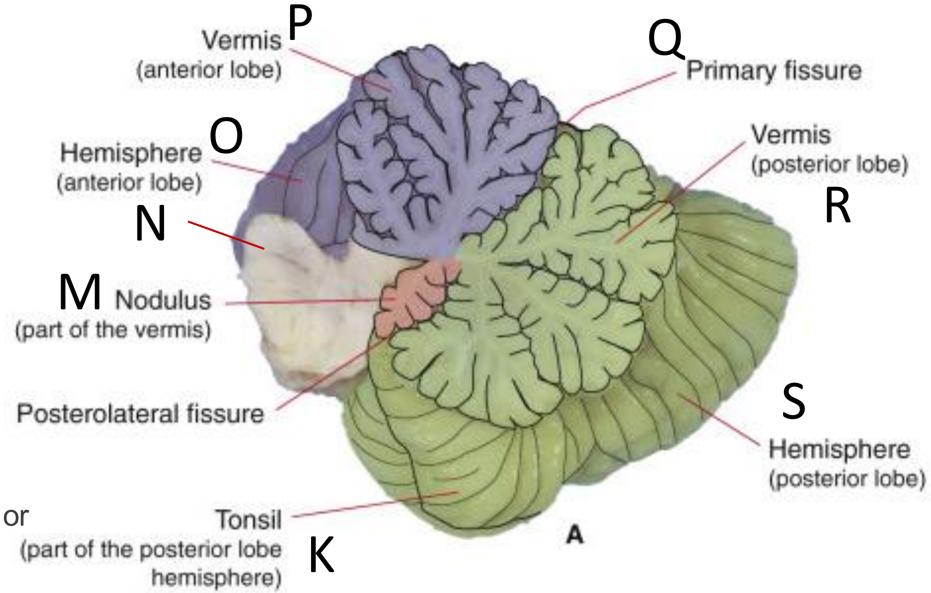


Pause here.....

Fig 8.20, Nolte's the Human Brain in Photographs and Diagrams, 5th Ed., 2020

Cerebellar Structures



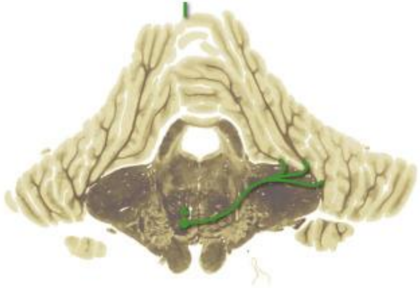


Cerebellar peduncles

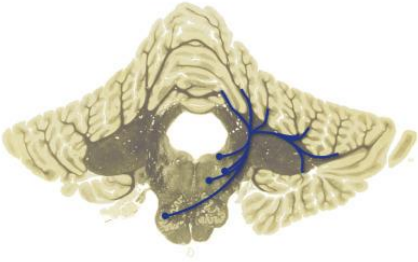
Info from? | Destination?

Which peduncle?

Which peduncle?

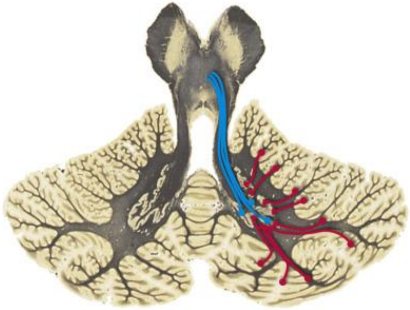
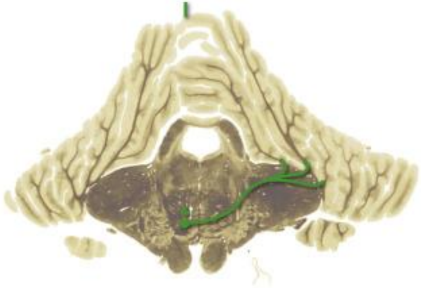
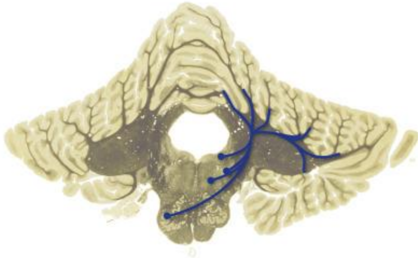


Which peduncle?



Pause here…..

Cerebellar peduncles

	Info from		Destination
Superior CP (SCP)	Cerebellar efferents: Cortex Purkinje cells → Dentate/ Interposed nuclei → Ipsi SCP		• Contralateral thalamus (VL) → motor/premotor cortex • Red nucleus
Middle CP (MCP)	Cerebellar afferents: Corticopontine tract → Pontine nuclei → Contra MCP (mossy fiber)		• Cerebellar cortex (mostly lateral/ medial part) → Purkinje cells → Dentate/interposed nuclei
Inferior CP (ICP)	Cerebellar afferents: 1. Vestibular nucleus/spinal cord → Vestibulo-/Spino-cerebellar tracts → Ipsi ICP (mossy fiber) 2. Inferior olivary nucleus → contra ICP (climbing fiber)		• Cerebellar cortex (vermis/flocculonodular /medial/lateral part) → Purkinje cells → Fastigial/ interposed/ dentate nuclei

Cerebellum Summary

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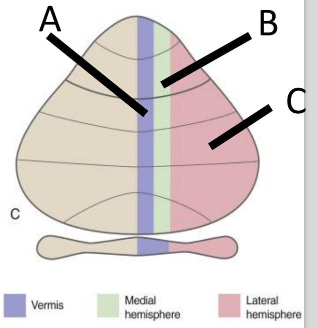
Input:

- From spinal cord/vestibular nuclei: ______________ side via _____________cerebellar peduncle at _________ level (smaller portion from the superior cerebellar peduncle)
- From cerebral cortex: ______________ side via ___________cerebellar peduncle at ________ level

Output:

From cerebellar cortex __________ cells → _________________ nuclei (_________, ______________, __________ nucleus) → _______________ side motor cortex, via _____________cerebellar peduncle at__________ level



Lesions:

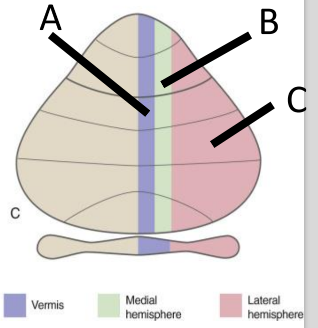
Lateral lesion (lateral cerebellar cortex): ________ataxia → fall toward ___________ side
Medial lesion (vermis, flocculonodular): ________ ataxia (postural instability, broad-based gait) → __________ (side) motor deficits
Damage to the flocculus also affects _____________________________

Cerebellum Summary

Input:

- From spinal cord/vestibular nuclei: ___ipsilateral___ side via ___inferior___ cerebellar peduncle at ___medulla___ level (smaller portion from the superior cerebellar peduncle)
- From cerebral cortex: ___contralateral___ side via ___middle___ cerebellar peduncle at ___pons___ level

Output:



inhibit

From cerebellar cortex ___Purkinje___ cells → ___Deep cerebellar___ nuclei (___dentate___, ___interposed___, ___fastigial___ nucleus) → ___contralateral___ side motor cortex, via ___superior___ cerebellar peduncle at ___midbrain___ level

Lesions:

Lateral lesion (lateral cerebellar cortex): ___limb___ ataxia – fall toward ___ipsilateral___ side

Medial lesion (vermis, flocculonodular): ___truncal___ ataxia (postural instability, broad-based gait) – ___bilateral___ motor deficits

Damage to the flocculus also affects ___eye movements (nystagmus)___

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Case 1

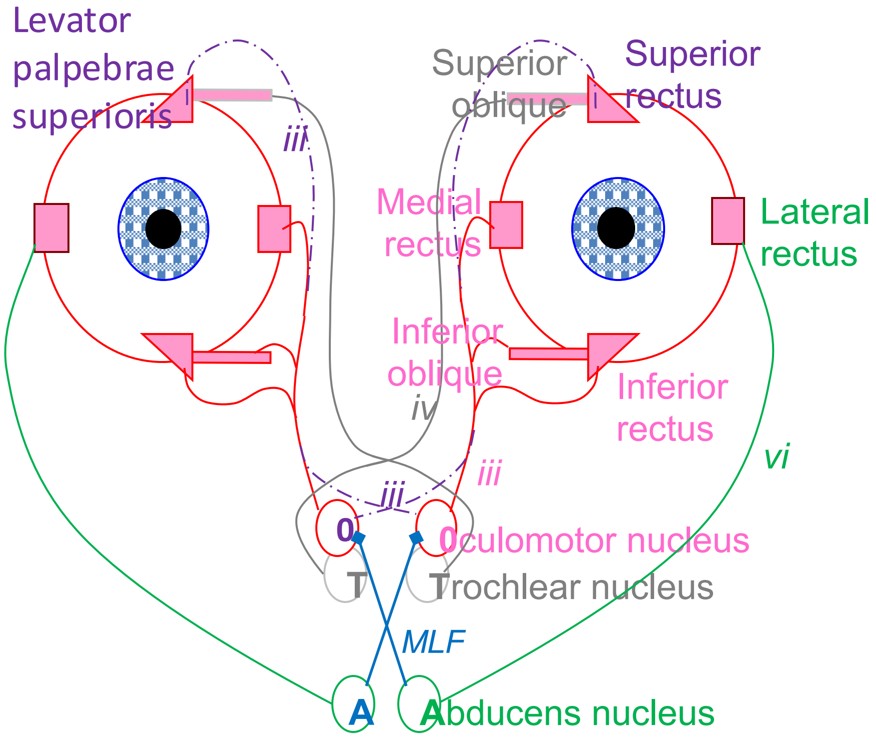
A 35-year-old female presents to the ED with complaints of diplopia and severe headache. History reveals these symptoms have progressed over several days. Physical examination reveals lateral strabismus, ptosis, and pupillary dilation for the left eye. Motor examination reveals normal strength and range of motion of upper and lower extremities. A CT scan of the head and orbits reveals an aneurysm. Which of the following is the most likely location for this aneurysm?

A) Anterior communicating artery
B) Left anterior cerebral artery
C) Left posterior communicating artery
D) Right posterior cerebral artery
E) Right posterior communicating artery

Damaged structure(s)?

Which side?

Eye Movement – CN III, IV, VI



A 35-year-old female presents to the ED with complaints of diplopia and severe headache. History reveals these symptoms have progressed over several days. Physical examination reveals lateral strabismus, ptosis, and pupillary dilation for the left eye. Motor examination reveals normal strength and range of motion of upper and lower extremities. A CT scan of the head and orbits reveals an aneurysm. Which of the following is the most likely location for this aneurysm?

Damaged structure(s)?

Which side?

Case 1

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A) Anterior communicating artery
B) Left anterior cerebral artery
C) Left posterior communicating artery
D) Right posterior cerebral artery
E) Right posterior communicating artery

Which artery?

Case 1

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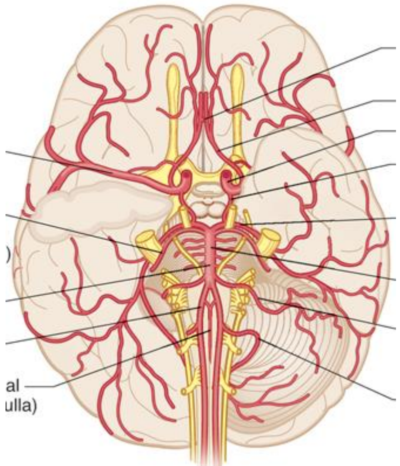

Left oculomotor nerve

Case 2

A 25-year-old man comes to the emergency department because he has been dizzy and nauseated and unstable while walking since undergoing a chiropractic treatment for neck pain several hours ago. Physical examination reveals absent pinprick and temperature sensation on his right face, left trunk, and limbs; ataxia of his right extremities; left-beating nystagmus; and hoarseness. Which of the following additional physical findings is most likely in this patient?

A) Deviation of the tongue to the right

B) Diminished vibratory sensation in the right extremities

C) Hyperreflexia of the left extremities

D) Inability to recognize the left side of his body

E) Right-sided ptosis and miosis

What structures were damaged?

At what level?

Which side?

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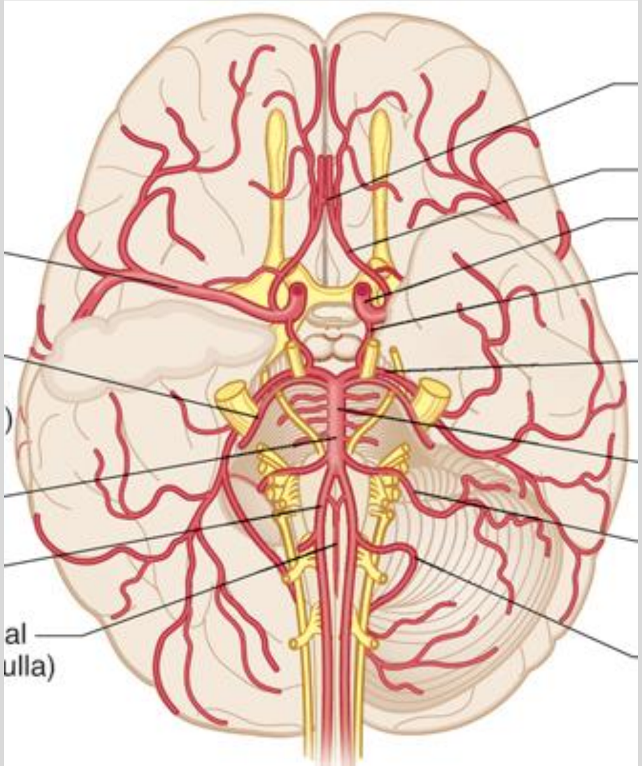

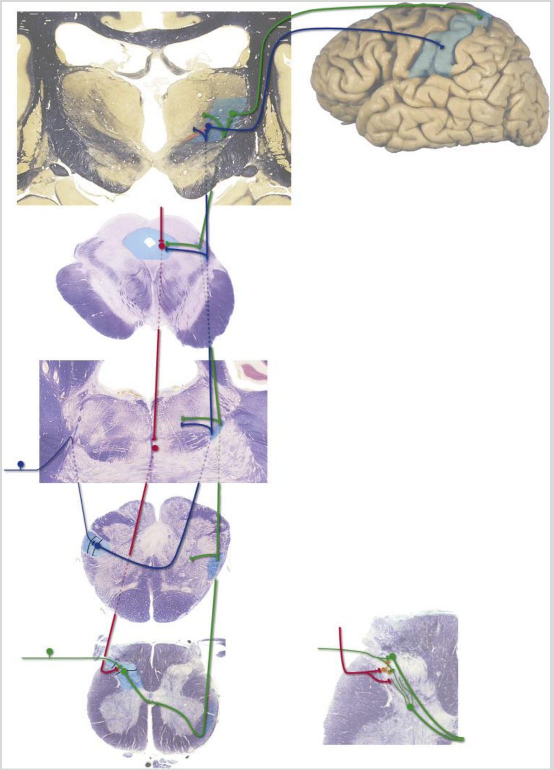

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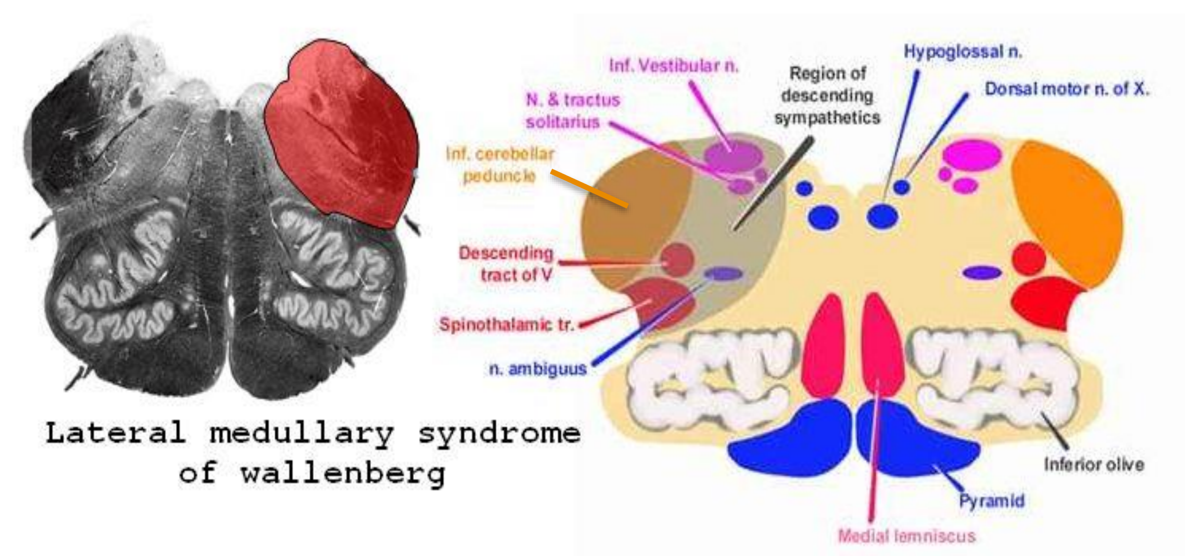

https://medchrome.com/basic-science/anatomy/lateral-medullary-syndrome-wallenberg-syndrome/

Case 2

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Horner's syndrome

Lateral medullary syndrome

A) Deviation of the tongue to the right

B) Diminished vibratory sensation in the right extremities

C) Hyperreflexia of the left extremities

D) Inability to recognize the left side of his body

E) Right-sided ptosis and miosis

Which blood vessel stroke is most likely causing it?

PICA or vertebral a. branches

Session Objectives

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- Review the CNS blood supplies.
- Differentiate the major functions of different lobes.
- Depict the major sensory & motor functional systems.
- Describe the major cerebellar pathways.
- Integrate the structure – function to identify lesions.

For review, please read:

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Nolte's the Human Brain in Photographs and Diagrams, 5th edition, 2020

Chapter 8. Functional Systems

Available at eClinicalKey:

https://www-clinicalkey-com.arktos.nyit.edu/#!/content/book/3-s2.0-B9780323598163000082

Ad - We will go over more practice items in the review session.

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Lecture and Lab Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R

THANK YOU!!

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